

AUCA · Introduction to Big Data Analytics

Week 1 Lab — Python Foundations

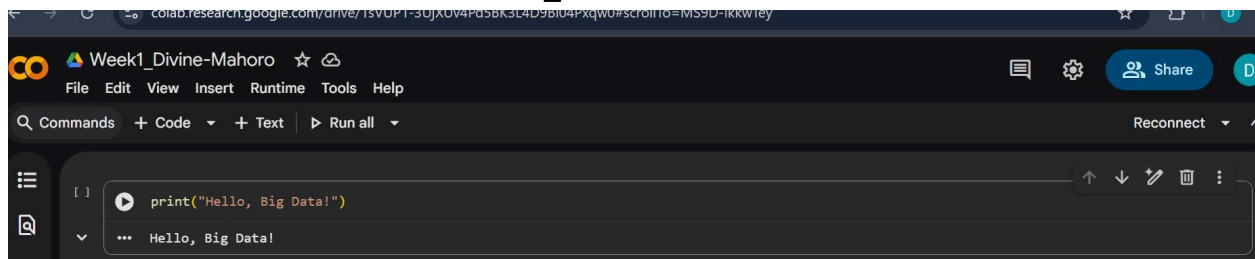
Names: Divine Mahoro

Student ID: 29109

Program: Software Engineering

Part 0 · Warm-Up (not graded)

Create a new notebook, rename it Week1_YourName, and run:

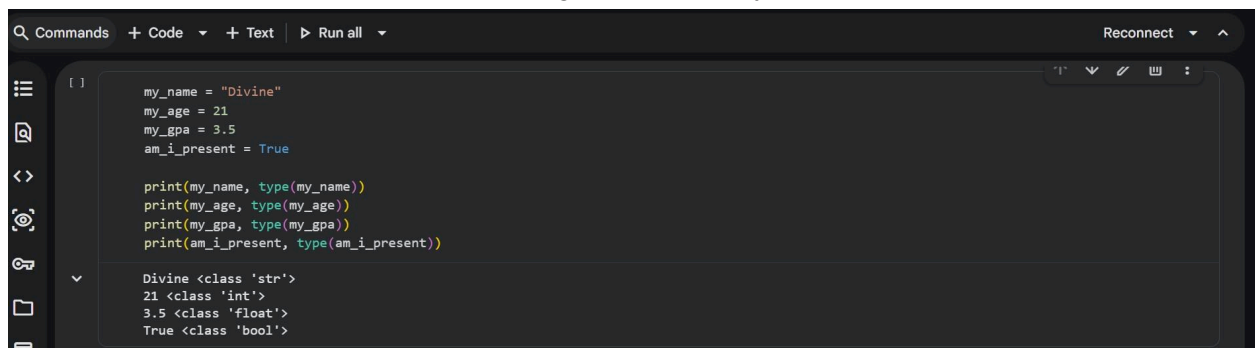


```
[ ] print("Hello, Big Data!")  
... Hello, Big Data!
```

Part 1 · Variables & Data Types (5 marks)

Exercise 1.1 (2 marks)

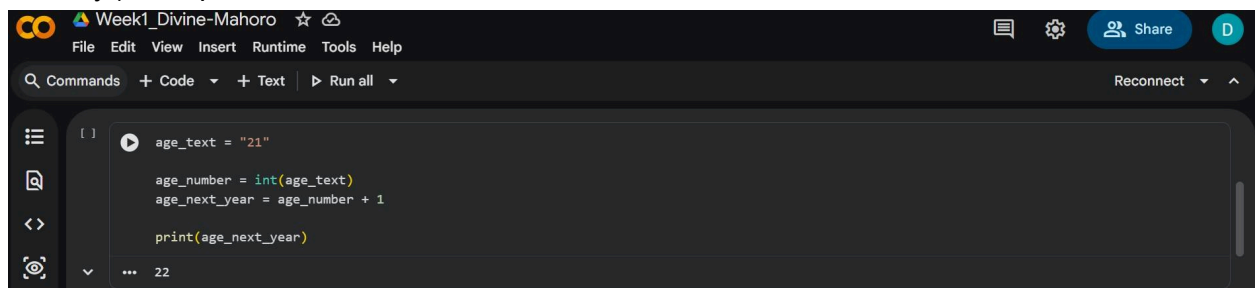
Create four variables about yourself: `my_name` (str), `my_age` (int), `my_gpa` (float — invent one), and `am_i_present` (bool). Print each variable together with its type.



```
[ ]  
my_name = "Divine"  
my_age = 21  
my_gpa = 3.5  
am_i_present = True  
  
print(my_name, type(my_name))  
print(my_age, type(my_age))  
print(my_gpa, type(my_gpa))  
print(am_i_present, type(am_i_present))  
  
Divine <class 'str'>  
21 <class 'int'>  
3.5 <class 'float'>  
True <class 'bool'>
```

Exercise 1.2 — Type conversion (2 marks)

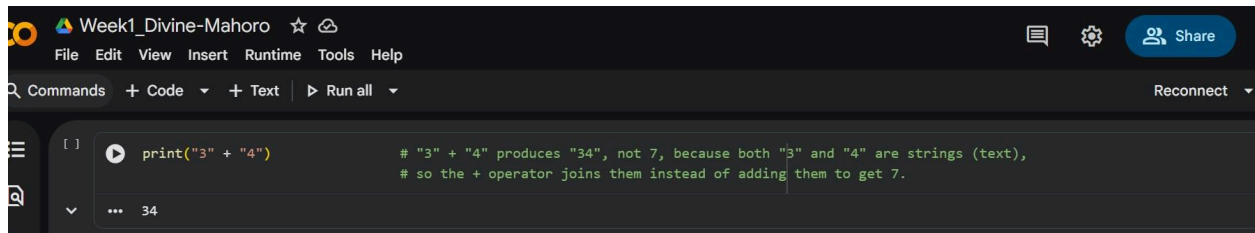
Real datasets often store numbers as text. Starting from the given line below, convert the value to an int, add 1 (a birthday!), and print the result.



```
[ ]  
age_text = "21"  
  
age_number = int(age_text)  
age_next_year = age_number + 1  
  
print(age_next_year)  
... 22
```

Question 1.3 — Think about it (1 mark)

What does "3" + "4" produce? Run it, then explain in one sentence why the answer is not 7.



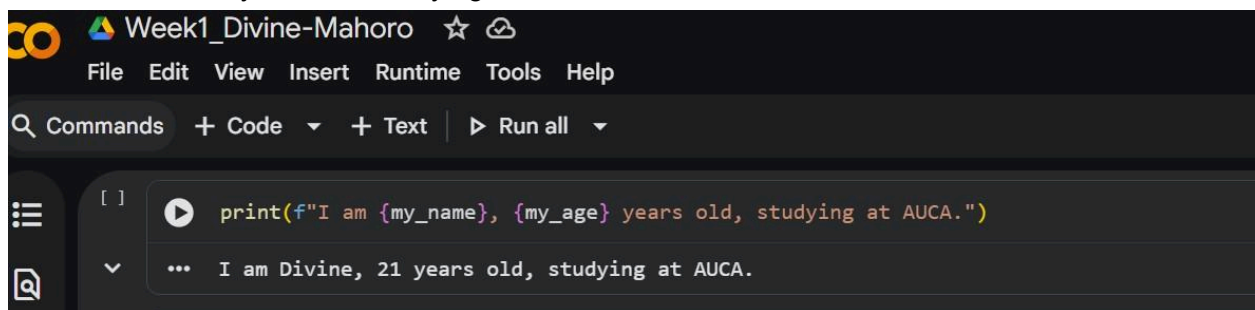
The screenshot shows a code editor window titled 'Week1_Divine-Mahoro'. The code being executed is `print("3" + "4")`. A comment explains that this produces "34" because both "3" and "4" are strings (text), so the + operator joins them instead of adding them to get 7. The output shown is `34`.

Part 2 · Strings — Working with Text (5 marks)

Exercise 2.1 — f-strings (2 marks)

Using your variables from Exercise 1.1, print the following sentence with one f-string:

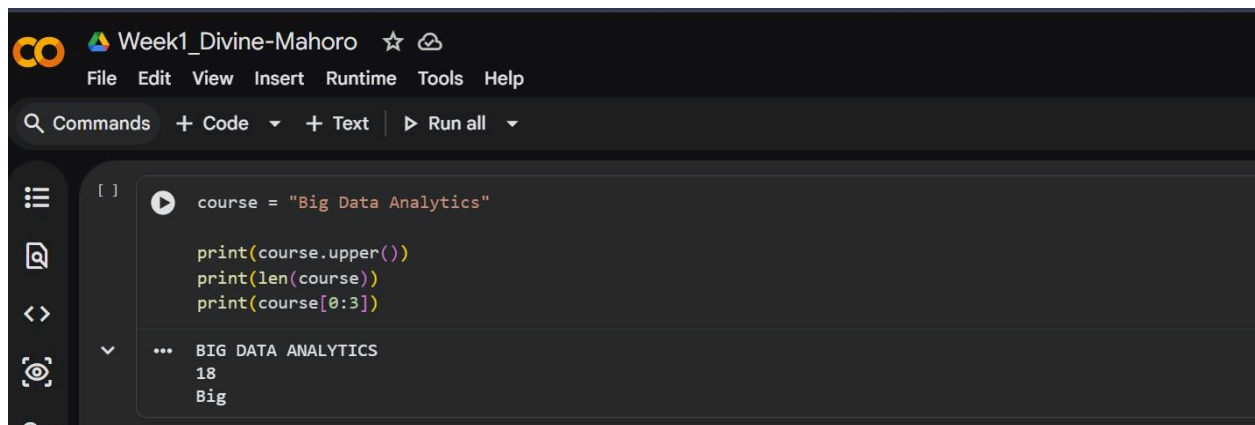
I am NAME, AGE years old, studying at AUCA.



The screenshot shows a code editor window titled 'Week1_Divine-Mahoro'. The code being executed is `print(f"I am {my_name}, {my_age} years old, studying at AUCA.")`. The output shown is `I am Divine, 21 years old, studying at AUCA.`

Exercise 2.2 — String tools (3 marks)

Given `course = "Big Data Analytics"`, print: (a) the course name in UPPERCASE, (b) the number of characters it contains, (c) only the first 3 characters.



The screenshot shows a code editor window titled 'Week1_Divine-Mahoro'. The code being executed is:

```
course = "Big Data Analytics"
print(course.upper())
print(len(course))
print(course[0:3])
```

The output shown is:

```
BIG DATA ANALYTICS
18
Big
```

Part 3 · Lists — Your First Descriptive Statistics (7 marks)

Exercise 3.1 (4 marks)

The quiz scores of your 5-person study group are given below. Print: (a) the first and the last score, (b) the number of scores, (c) the average, (d) the highest and the lowest score.

The screenshot shows a code editor with a dark theme. The top bar includes the logo, the name 'Week1_Divine-Mahoro', and icons for star and cloud. Below the bar is a menu with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar with 'Commands' and buttons for '+ Code', '+ Text', and 'Run all' are visible. The main editor area contains a Python script:

```
scores = [72, 85, 91, 64, 78]

print(scores[0], scores[-1])
print(len(scores))
print(sum(scores) / len(scores))
print(max(scores), min(scores))
```

 The output console on the right shows the results:

```
72 78
5
78.0
91 64
```

Exercise 3.2 — Lists change (3 marks)

A new member joins with a score of 88, and you discover the 4th score (position 3) should be 66, not 64. (a) Append

88 to the list, (b) replace position 3 with 66, (c) print the updated list and its new average.

The screenshot shows the same code editor as before. The Python script is updated to:

```
scores.append(88)
scores[3] = 66

print(scores)
print(sum(scores) / len(scores))
```

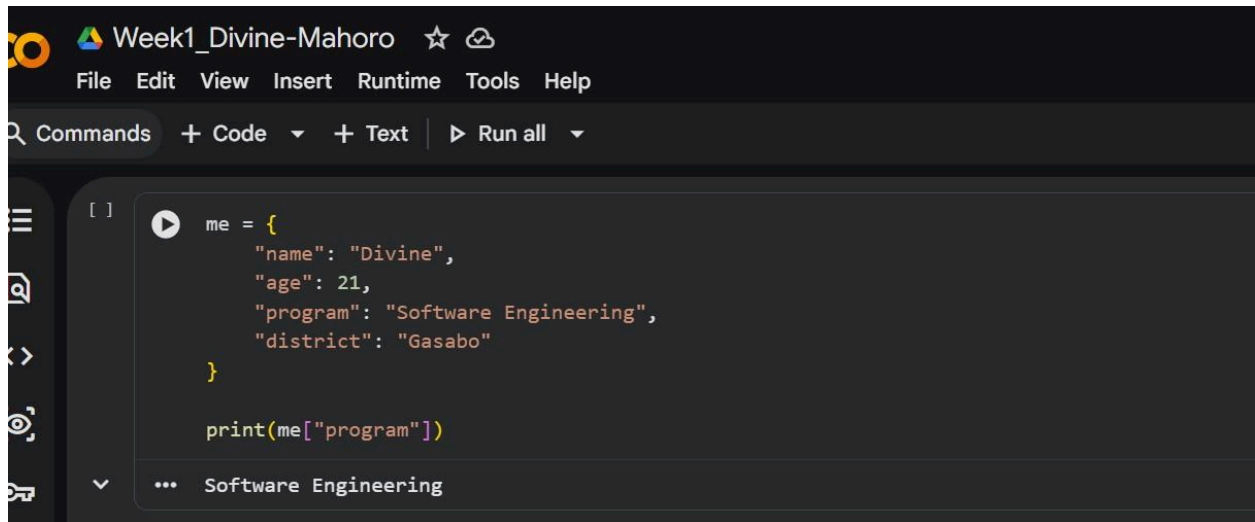
 The output console now shows:

```
[72, 85, 91, 66, 78, 88]
80.0
```

Part 4 · Dictionaries — Labeled Data (5 marks)

Exercise 4.1 (2 marks)

Build a dictionary called `me` describing yourself with the keys "name", "age", "program", "district". Print your program using `me["program"]`.



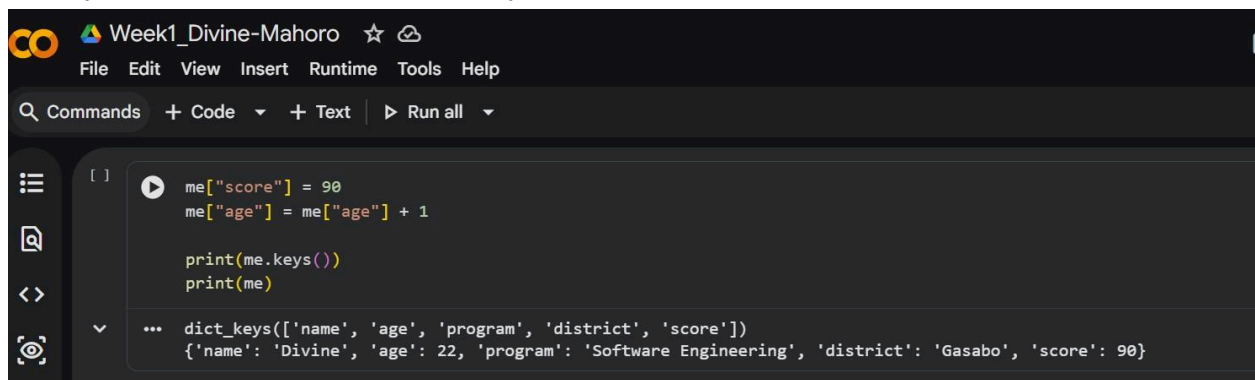
The screenshot shows a code editor window titled "Week1_Divine-Mahoro". The menu bar includes "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu bar, there is a search bar and buttons for "Commands", "+ Code", "+ Text", and "Run all". The code editor displays the following Python code:

```
[ ] me = {  
    "name": "Divine",  
    "age": 21,  
    "program": "Software Engineering",  
    "district": "Gasabo"  
}  
  
print(me["program"])
```

The output of the code is displayed below the editor: "... Software Engineering".

Exercise 4.2 — Update and extend (3 marks)

(a) Add a new key "score" with a value of your choice, (b) increase "age" by 1, (c) print all the keys with `me.keys()` and then the whole dictionary.



The screenshot shows the same code editor window. The code has been updated to:

```
[ ] me["score"] = 90  
me["age"] = me["age"] + 1  
  
print(me.keys())  
print(me)
```

The output of the code is displayed below the editor: "... dict_keys(['name', 'age', 'program', 'district', 'score'])
{'name': 'Divine', 'age': 22, 'program': 'Software Engineering', 'district': 'Gasabo', 'score': 90}"